



Online collaborative problem-solving as a tangible outcome of digital skills in technical and vocational higher education

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ABSTRACT

In today's world, specific skills are essential for success in work and life. Educators, governments, and industry leaders recognize collaborative problem-solving (CPS) and digital skills as necessary for the 21st century. However, our current understanding of these skills' interconnection and development within technical and vocational higher education remains limited. This study aims to achieve two main objectives. First, we want to examine the level of digital skills among technical and vocational higher education students in Chile and how it relates to their socio-economic background. Second, we aim to investigate how students' different levels of digital skills influence their satisfaction with online collaboration and their assessment of group processes after completing an online CPS task. The study identified three distinct groups of students based on their digital skills: high-skilled, medium-skilled, and low-skilled. The low-skilled group faced more socioeconomic disadvantages compared to the other two groups. Moreover, students with low digital skills reported lower satisfaction with online group work and a poorer self-evaluation of the group processes involved in the collaborative online task. Insights for technical and vocational higher education institutions to foster targeted support of digital skills development for their student body are discussed.

1. Introduction

Preparing students to face the current technology-driven job market is one of the most critical challenges for technical and vocational higher education (TVHE). To successfully navigate the world of work, it is crucial for students to effectively develop problem-solving skills, communication and collaboration with others, and digital media literacy. Students with less training in this so-called 21st-century skills will struggle to fit in their future workplace (UNESCO, 2022; UNESCO-UNEVOC, 2020). Future occupations increasingly require the ability to work in groups, and problem solving skills are currently central to effectively completing everyday job tasks and challenges (Fajaryati et al., 2020; González-Pérez & Ramírez-Montoya, 2022; Goulart et al., 2022; Szabo et al., 2020; Vista, 2020). Despite its importance, little is known about the relevant factors for effectively developing collaborative problem-solving skills (CPS) (Graesser et al., 2018), particularly in schools and other educational contexts (Koehler et al., 2022; Rosen et al., 2020).

21st-century skills encompass abilities, competencies, knowledge, and learning approaches identified as critical for succeeding in

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work and life. Both CPS and digital skills have been acknowledged as 21st-century skills by educators, governments, and industry leaders (Van Laar et al., 2017; World Bank, 2018); however, little is known about their relationship or how to develop these skills in TVHE (Van Laar et al., 2020).

The purpose of this study is twofold. First, to examine the level of digital skills among TVHE students in the Chilean context and its relationship with their socioeconomic background. Second, to explore the relationship between students' different levels of digital skills and both i) their satisfaction with working collaboratively online and ii) their self-assessment of group processes after completing an online CPS task.

2. Literature review

2.1. Digital skills and the digital divide

Digital literacy can be conceptualized as the ability to use information and communication technologies (ICTs) in ways that help individuals to achieve tangible and valuable outcomes in everyday life (Helsper, 2012; Helsper et al., 2015; Van Deursen et al., 2017). Tangible outcomes refer to the concrete, measurable, and valuable benefits that individuals gain through their adept use of ICTs in their daily lives (Van Deursen et al., 2017). Tangible outcomes encompass, for instance, improved access to information, enhanced communication capabilities, expanded opportunities for learning and education, increased efficiency and satisfaction in performing tasks, and more generally, the ability to actively engage in modern society and the digital economy (Helsper et al., 2015; Livingstone et al., 2023).

Commonly, researchers and policymakers have taken a skills approach to operationalize digital literacy and, thus, inform study measurements and design interventions to enhance digital literacy. Digital skills transcend technical competencies to accessing and using ICTs and encompass all skills related to problem-solving in a digital environment, such as searching, assessing, summarizing, analyzing, representing, communicating, creating, and sharing information (Claro et al., 2012, 2015; Fraillon, 2012; Fraillon et al., 2013). Furthermore, the social media rise over the last decade has placed social skills such as communication, social content creation, and collaboration with others at the center of the conception of digital literacy (Calvani et al., 2012; Van Deursen et al., 2014; Van Dijk, 2017, 2020).

In the education arena, research has focused on the inequalities among students in digital skills, the so-called “digital divide” (Cabello & Claro, 2017; Claro et al., 2015; Van Deursen & Van Dijk, 2014, 2019; Van Dijk, 2013, 2017). In the origins of the concept, the digital divide referred almost exclusively to differences in access to ICTs among different populations due to financial or resources disparities (Hargittai & Hinnant, 2008; Selwyn, 2004; Van Dijk, 2020). As a result, numerous policies aiming to reduce this gap have been implemented, often based on equipping educational institutions with ICT resources, such as computers, laptops, tablets, educational software, and Internet (e.g., Näslund-Hadley et al., 2009; Severin & Capota, 2011).

Nevertheless, current evidence points out that as the ICT access gap reduces significantly, other inequalities rapidly emerge related to the manners in which the already available technology is being used and the differences in the capacity among individuals to take advantage of ICTs to obtain beneficial outcomes (Elena-Bucea et al., 2021; Jara et al., 2015; Liu, 2021; Van De Werfhorst et al., 2022). Therefore, disparities in digital skills include more than just financial aspects; they encompass cognitive, sociodemographic, and cultural differences, giving rise to a phenomenon known as the “second digital divide” (Attewell, 2001; Lythreitis et al., 2022; Scheerder et al., 2017; Zhao & Elesh, 2007). This enhanced understanding of the digital divide concept expands the issue from a singular focus on access to a broader, multidimensional problem (including social and cultural factors) (Elena-Bucea et al., 2021).

Research in many cultural contexts shows that students from disadvantaged socioeconomic backgrounds present difficulties in accessing and taking advantage of ICTs (Elena-Bucea et al., 2021; Liu, 2021; Van De Werfhorst et al., 2022). Similarly, Claro et al. (2015) provide evidence that helps understand the digital divide in Chile, a country with substantial social inequalities that are reflected in a highly class-segmented educational system (Dupriez et al., 2023). The authors found that the socioeconomic background of high school students significantly predicted performance in a digital skill standardized test. Furthermore, they showed that the influence of socioeconomic background on Chilean students' performance on a digital test was more significant than the effect on other traditional tests (mathematics and language), suggesting that the digital domain could amplify already existing skill gaps.

2.2. Collaborative problem-solving (CPS)

CPS, broadly defined, is the ability of an individual to engage effectively in a process through which two or more agents attempt to solve a problem by sharing the understanding and effort required to reach a solution (Graesser et al., 2018).

CPS is an intrinsically complex mechanism that integrates cognition components in individual problem-solving and collaboration. Conceptually, the cognitive dimension of individual problem-solving includes the understanding and representation of problem content, the application of problem-solving strategies, and the implementation of self-regulatory and metacognitive processes to monitor goal-oriented progress (Funke et al., 2018; Graesser et al., 2018; Sun et al., 2020). A problem-solving task stands apart from a routinary task by demanding more than straightforward execution of instructions or routine actions (Rahman, 2019). Problem-solving tasks encompass navigating complexities, analyzing multifaceted information, engaging in critical and creative thinking, and summarizing and interpreting data to craft an adequately response to the task (Di et al., 2019). Examples of problem-solving include tasks with multiple correct solutions and diverse pathways leading to a satisfactory response, as well as those that involve delving into complex or controversial topics, where exploring different perspectives and constructing well-reasoned arguments are integral (Treffinger et al., 2023).

On the other hand, collaboration factors require broader cognitive and social skills that enable understanding of the team's shared goals, knowledge and information flow, team organization building, trust building, and coordination of actions around group collective objectives (Funke et al., 2018; Graesser et al., 2018; Sun et al., 2020). Interdependence in the design of a CPS task is important to elicit more profound collaboration (Scoular et al., 2020). Interdependence refers to the reliance or interconnectedness among individuals that is required to solve a problem. In other words, a collaborative task is interdependent if its successful resolution would be difficult to achieve by just one member of the group, and thus, requires relying on others' inputs, roles, expertise, or actions. Providing differentiated resources to individuals to solve a problem, assigning distinct roles within the group, or forming teams whose members possess diverse skills or expertise are examples of ways to imprint interdependence to the design of a task (Scoular et al., 2020).

Empirically, the PISA test applied in 2015 sheds light on CPS in the educational context including the collaborative domain by incorporating the theoretical bases of individual and group cognition (OECD, 2017). Results were released in 2017, rendering this study the most recent international and large-scale assessment of CPS.¹ The CPS PISA test was administered to 15-year-old students from more than 50 countries. The test employed a computer-mediated environment in which students engaged with virtual agents to assess and evaluate their CPS skills within digital problem-solving scenarios. The study found that only a small share of students (28%) could solve basic collaborative problems concluding that there are substantial challenges in effectively developing these skills (OECD, 2017). Chilean students scored 457 points, significantly below the OECD average of 500, and ranked 38th out of 52 countries (OECD, 2017).

In Chile, there has been an emergent exploration of CPS in education within the literature. For example, Saadati & Felmer (2021) examined the impact of a year-long teacher professional development that promoted the use of CPS activities in the classroom. Using a quasi-experimental design, the study compared the problem-solving performance of 6th grade students and found a statistically significant difference in favor of the group of students whose teachers applied CPS in class versus those whose teachers applied traditional methods. On the other hand, Rojas et al. (2022) opens an avenue for assessing the use of novel automatization technologies in the design of CPS activities. They developed a feedback system that uses an automated monitoring system aimed to enhance fifth-grade student communication during a CPS activity. The activity involved a videogame especially designed to be used in an educational setting. They found that feedback allowed the students to perform better in the game, improve their communication during the activity, and further develop their CPS skills. While significant, none of these studies assessed the link between digital skills and CPS, particularly within the context of technical and vocational higher education.

CPS skills are expected to improve individuals' career readiness directly and, in so doing, benefit society at large (Fiore et al., 2018; Hesse et al., 2015; OECD, 2017; Wiltshire et al., 2018). In this sense, educational institutions are key actors in developing students' CPS skills because the academic environment provides regular opportunities to enroot CPS pedagogies and methodologies, thereby progressing social skills and positive attitudes towards collaboration in general and CPS in particular (OECD, 2017). Despite its significance, students' level of CPS skills is generally low, efforts or initiatives for CPS development are rarely assessed, and there is little clarity on the curricular strategies and proposals that work in educational contexts for CPS training (OECD, 2017).

The digital transformation, which has been underway for decades but was accelerated due to the COVID-19 pandemic, adds a new layer to CPS skills' advancement and attainment. Digital technologies are now essential for group activities in work and education. Amidst this scenario, assessing the mediating role of digital skills in CPS development has become urgent.

CPS and digital skills have been acknowledged as 21st-century skills (Fiore et al., 2018; National Research Council, 2011; Van Laar et al., 2020). However, little is known about their relationship or how to properly develop these skills in formal education. Moreover, the existing body of knowledge is not particularly adapted to technical higher education or the context of developing countries. The present study aims to fill these literature gaps by examining some fundamental aspects of CPS and digital skills' role in Chile's TVHE. More specifically, this study assesses whether there is evidence of a digital divide among TVHE students. In addition, it examines the relationship between these students' digital skills and their satisfaction with and self-assessment of an online CPS task.

3. Conceptual framework: digital skills to tangible outcomes

This study relies on the perspectives of Helsper et al. (2015), whose conceptual framework allows to conceptualize and provide structure to the relationship between digital skills and individual outcomes. Employing this model as the conceptual framework of this study enables us to delineate and organize the relationship we are focusing on, namely, the connection between students' different levels of digital skills and the following outcomes: i) satisfaction with working collaboratively online and ii) self-assessment of group processes after completing an online CPS task.

One of the fundamental premises of the model is that engagement with or uses of ICTs should not be viewed as the goal of researchers and policymakers, but rather, whether these engagements or uses translate into tangible and measurable benefits for individuals in different spheres of their everyday life. According to the model, lower level of digital skills lead to lower levels of engagement with the Internet, resulting in a lower likelihood for an individual to achieve tangible outcomes in modern society (Van Deursen et al., 2017).

The model begins by clearly defining and separating the activities individuals undertake in the digital sphere (i.e., digital resource fields) from the tangible outcomes in different domains of everyday life (i.e., offline resource fields) derived from said digital engagement. Translating engagement with ICT into tangible outcomes depends on four categories of digital skills: Operational,

¹ More recent versions of PISA test do not include a CPS assessment module.

Information and Navigation, Social Communication, and Content Creation skills. Operational and Information Navigation skills are preconditions for Social and Creative skills, which are more advanced. These skills are transferable (i.e., transversal, not specific to one online activity or task) and required for an individual to perform any online task successfully. Insufficient skills in one or more of these domains may limit accomplishment or efficiency when undertaking an online task and, consequently, may cause the individual to fail in turning the activity or task into a desired tangible outcome (Van Deursen et al., 2017).

In addition to digital skills, the ability to translate engagement with ICT into tangible outcomes also depends on the individual's socioeconomic, cultural, and personal resources in what is called a "corresponding fields model" (Van Deursen et al., 2017).

3.1. Corresponding fields model

The model recognizes a direct correspondence between the online and the offline sphere in key resource fields: economic, cultural, social, and personal (Helsper, 2012) (see Fig. 1). For example, resources in the *economic offline* field consist of education, employment, and income and finances. Regarding education, this offline resource could be operationalized as any educational achievement (e.g., formal training or a degree). Equivalently, the corresponding *economic online* field consists of financial and commercial digital resources and online information and learning, which could be operationalized as online banking activities, e-learning, or online course completion. Each separate field contains a pool of interconnected offline and online resources, and each of these resources can be identified, operationalized, and measured. The corresponding fields model postulates that the social (offline) and the digital (online) world are directly connected, and that this social-digital link is stronger between corresponding fields of offline and online resources. This study focusses its attention on the Economic field and the Education subfield.

3.2. Tangible outcomes in the economic field

To understand the importance of digital skills, we need to focus on the tangible or real outcomes its use can generate in individuals' everyday lives (Helsper et al., 2015; Helsper & Eynon, 2015; Van Deursen et al., 2017). Such a focus requires classification and operationalization (i.e., measures) of benefits resulting from ICTs use. The economic field includes education, employment, and income and finances subfields. Digital skills may improve the economic outcomes of an individual in multiple ways. For example, a person may find a job because of immediate access to a broader range of job opportunities through the Internet or obtain performance rewards in their company for completing an online certification.

Fig. 2 shows the flow of the model from digital skills to tangible outcomes. Let us illustrate this with an example from the education subfield: a person, using their digital skills for looking for information and navigating the Internet (Information Navigation) may register a course or program thanks to the information they found online (Internet use) and get an online certificate (Tangible Outcome). It is worth noting that, according to the model, satisfaction with the online course or program (i.e., the intrinsic enjoyment elicited from participating in online educational activities) is also considered a tangible outcome.

The model identifies four critical digital skills: Two elemental -Operational, and Navigation Information- and two advanced -Social and Creative. Operational skills refer to the technical ability to effectively use digital devices and software (e.g., how to operate computers, and software applications such as word processors, and spreadsheets). Navigation Information skills involve the ability to find, evaluate, and use information effectively on the internet (e.g., conducting online searches, assessing the credibility of sources, and navigating websites to locate relevant information). These two skills are elemental in the sense that they serve as the foundational blocks upon which the more advanced skills in the model are developed (Helsper, 2015; van Deursen, Helsper, & Eynon, 2014; van Deursen & van Dijk, 2014). Social skills refer to the ability to communicate, collaborate, and interact with others effectively through digital channels (e.g., using social media platforms, online messaging services, video conferencing tools, and other digital communication technologies to connect with peers, colleagues, and online communities). Lastly, creative skills involve the ability to generate new content using digital tools and platforms (e.g., creating multimedia content such as posts, videos, graphics, and websites).

Previous research highlights that operational skills are particularly susceptible to rapid changes due to the ongoing and accelerated advancement of technical tools (Ala-Mutka, 2011; Iordache et al., 2017). Operational skills are more likely to be acquired through

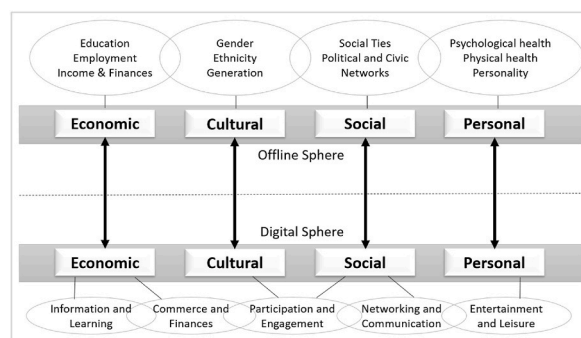


Fig. 1. Conceptual model of the links between offline and digital fields.

Source: Helsper, 2012; Helsper et al., 2015.

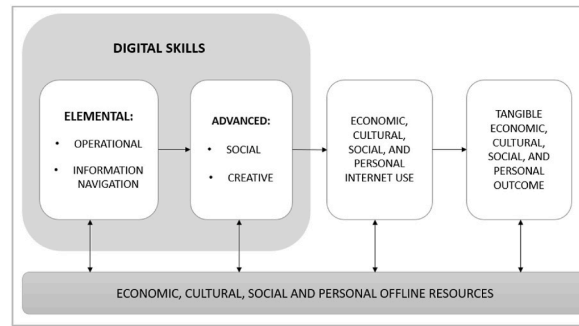


Fig. 2. Model of the relationship between digital skills and individuals' outcomes.

Source: [Helsper \(2015\)](#).

more formal training, as they are frequently integrated into computer or digital literacy programs or courses ([Iordache et al., 2017](#)). Conversely, the rest of the skills -especially in the Social and Creative domains- are commonly developed through self-directed learning and hands-on experience in real-world contexts ([Ferrari, 2013](#); [Hargittai, 2007](#); [Iordache et al., 2017](#)). As such, the development of these skills is likely influenced by individual experiences, interests, and opportunities for social interaction and creative expression ([Ferrari, 2013](#); [Hargittai, 2007](#); [Iordache et al., 2017](#)). In the model, the background of the individual, which is specified as their economic, cultural, social, and personal offline resources affect the chain of events from beginning to end (e.g., socioeconomic background influences digital skill levels). But, at the same time, any changes in skill levels, the use of Internet or the tangible outcome achieved can have an effect on the individual offline set of resources as well.

This conceptual framework by unraveling the pathway from digital skills to tangible outcomes offers an enlightening approach to study CPS in the educational context. Applying the model's logic to the examination of CPS unfolds as follows: Leveraging their elemental skills -Operational and Information Navigation digital skills- students can navigate through online sources, and be able to gather, assess, and synthesize data. Advanced digital skills such as Social and Creative enable them to sustain online group communication and use technology in creative ways in order to present a solution to the online problem. Lastly, the resulting tangible outcomes extend beyond the mere completion of the online CPS task; they also encompass the acquisition of novel insights, problem-solving strategies, and, importantly, the satisfaction derived from participating in the CPS task.

4. This study

This article aims to contribute to the scarce literature examining digital skills and CPS in higher technical education. The purpose is twofold. First, to assess whether there is evidence of a digital divide regarding skills among TVHE students in the Chilean context. Second, to examine the relationship between students' different levels of digital skills and both i) their satisfaction with working collaboratively online and ii) their self-assessment of group processes after completing an online CPS task.

Conceptually, this study draws from the literature concerning digital skills and CPS. We examine the connection between these two through the lens of [Helsper et al. \(2015\)](#) by positioning satisfaction with and self-evaluation of and an online CPS task as tangible outcomes derived from an effective use and engagement with ICTs.

This study specifically examines tangible outcomes in the economic field, with a particular focus on the subfield of education. We hypothesize that the students' digital skill level as well as their socioeconomic, cultural, and personal background influence their ability to engage and complete an online CPS task. According to Helsper, Van Deursen, and Eynon's conceptualization, this achievement should be conceived as a tangible economic outcome for the individual, in the subfield of formal education.

The study research questions are as follows:

Digital skills:

- 1) Can different students' profiles (clusters) be identified according to students' digital skills? How do these profiles vary across the four digital skills dimensions: Operational, Information Navigation, Social, and Creative?
- 2) How do students' socioeconomic background relate to the previously identified digital skills profiles? Is there evidence of a digital divide?

From digital skills to tangible outcomes:

- 3) To what extent do digital skills profiles (cluster membership) predict satisfaction with working collaboratively on an online CPS task? Which additional factors of the student's socioeconomic background influence this satisfaction?
- 4) To what extent do digital skills profiles (cluster membership) predict the evaluation of the online CPS task? Which additional factors of the student's socioeconomic background influence this evaluation?

Emerging from the research question, we outline the following hypotheses for this study:

Hypothesis related to Research Question 1:

H1. Different clusters of students based on digital skills are identified across the Operational, Information Navigation, Social, and Creative dimensions, indicating distinct profiles of digital skill sets among students.

Hypothesis related to Research Question 2:

H2. Socioeconomic background is significantly related to the identified digital skills profiles, revealing variations in digital skills across different socioeconomic levels and providing evidence of a digital divide among students.

Hypothesis related to Research Question 3:

H3.1. Profiles with higher digital skills significantly predict higher levels of satisfaction with collaborative work in the online CPS task.

H3.2. Specific socioeconomic background factors influence satisfaction levels.

Hypothesis related to Research Question 4:

H4.1. Profiles with higher digital skills significantly predict better evaluations of the online CPS task.

H4.2. Specific socioeconomic background factors influence evaluations.

5. Methods

5.1. Context and research design

The study takes place in a prominent TVHE institution in Chile offering a wide range of programs, including engineering, technology, business, health, hospitality, and culinary arts. It operates across multiple campuses, serving students from diverse backgrounds and geographic regions of the country. The study was conducted during the second semester of 2021 amidst the COVID-19 pandemic. Thus, despite the institution traditionally conducting mostly in-person classes, some courses transitioned to online instruction that year.

The present study uses a quantitative non-experimental design to examine students' different profiles regarding their digital skills and the relationship between these profiles and both: i) students' satisfaction with working in groups online, and ii) students' self-assessment of group processes while completing an online CPS task which is a requirement to get course credits. The study involved collecting data on students' digital skills and implementing an online CPS task as part of the online course "Effective Communication". This course is offered each semester and is a common first-year requirement for all students independently of the selected career path.

For the CPS task, students are asked to collaboratively formulate a response to a contested statement made by an economist about global warming. The goal is to collectively -in groups of 4 or 5 students- draft a response text representing the group's viewpoints on the issue (see detailed CPS task description in [Appendix A](#)). This CPS task delves into a contentious matter demanding critical thinking, analysis, assessment of diverse viewpoints within the group, synthesis of ideas, and ultimately, the construction of a cohesive argument. These are essential facets of problem-solving (Di Leo et al., 2019; Treffinger, 2023).

The study was conducted in three stages:

- a) Stage 1 (pre-CPS task): Carried out in week 10 of the course, at the beginning of the online session, right before implementing the online CPS task. Students were asked to answer a questionnaire with two main modules. The first asked about students' socioeconomic characteristics, and the second about their self-reported digital skills.
- b) Stage 2 (CPS task implementation): A problem was presented to the students in week 10 of the course to be solved through collaborative work in groups of 4–5 individuals. The group had to work asynchronously to solve the task and had a two-week deadline to submit their final responses (see task details and materials in [Appendix A](#)).
- c) Stage 3 (post-CPS task): This stage was carried out in week 12, at the end of the session. Students were asked to answer two online questionnaires: the first asked about students' satisfaction with working in groups, and the second collected information about students' self-assessment of group processes while completing the online CPS project.

The implementation of the CPS task and the application of the questionnaires were carried out online by the teachers of the "Effective Communication" course. For this, 95 teachers were trained by the research team during twelve meetings via videoconference. During the training, the CPS task material and detailed instructions for applying the CPS task in class was presented to the teachers (see activity description and learning materials in [Appendix A](#)).

5.2. Sample

We analyzed the data of a sample of 939 valid cases within the cohort enrolled in the "Effective Communication" course. Valid cases comprised students who i) participated and completed the CPS online task and ii) answered our set of three questionnaires (specified in the following section). Our response rate was just above 70%. Students were distributed on 27 campuses across the 15 regions of Chile. Participants belonged to 31 different programs, which were grouped into three areas: Management, Hospitality, and Tourism; Engineering; and Health. The sample distribution can be found in [Table 5](#).

5.3. Instruments and measures

Three self-reported questionnaires were applied to the students. The first, at the initial stage (pre-task) to measure students' digital skills and socioeconomic characteristics, and two after the task implementation, aimed at assessing students' satisfaction with working in groups and evaluation of group processes while working in the online CPS task.

5.3.1. Digital sphere: from digital skills to tangible outcomes questionnaire (DISTO)

To measure digital skills, we selected the self-reported questionnaire From Digital Skills to Tangible Outcomes (DISTO) developed by Van Deursen et al. (2014). This instrument measures individual's digital knowledge, engagement, and ICT use, and map these to four dimensions of digital skills: Operational, Navigation Information, Social, and Creative skills, aligning with our conceptual framework. This instrument has been validated, adapted, and used in large and relevant projects about digital skills in different countries and regions of the world,² and has been adapted to the Chilean population throughout a rigorous process that demonstrated its validity and reliability in this particular context³ (Claro, 2017).

We decided to use a self-reported questionnaire instead of a performance assessment to measure digital skills, despite the limitations associated with this type of instrument (e.g., social-desirability and individual self-beliefs bias) (Rosman et al., 2015). This is due to mainly two factors. First, there was a specific request from the TVHE institution to avoid overloading students with evaluative tasks during the pandemic. Important concerns were raised about both potential low response rates and the stress that a digital skills performance assessment could cause for students. Second, there were logistical challenges such as administrative, coordination, and budgetary constraints associated with implementing a performance assessment for a large sample of students across the different regions of the country. Thus, opting for a performance assessment would have meant compromising both sample size and geographic coverage.

Table 1 summarizes the main characteristics of the DISTO questionnaire. Regarding our sample, DISTO presented good reliability and appropriate levels for fit indices (see Appendix B).

5.3.2. Tangible outcomes: satisfaction with working in groups (SGP) and assessment of CPS task (CPS)

According to Helsper et al. (2015) "to really understand tangible outcomes both the quantitative and qualitative elements need to be included in the measurements." (p.15). Navigating successfully through group collaboration processes to accomplish a task does not directly imply that the individual felt gratification or enjoyment. For example, an individual may strongly agree that the group found a collaborative way of working to solve the problem (quantity) but, at the same time, felt very dissatisfied with how the group communicated about the task (quality). Thus, we decided to measure tangible outcomes in two ways: quality (satisfaction with group processes) and quantity (assessment of CPS processes) of the outcome. Quality statements were measured with satisfaction scales and quantity using agreement scales (see Table 2).

Measuring Online CPS Quality (satisfaction): SGP. The SGP questionnaire was developed using the framework created by Han & Beyerlein (2016). This instrument measures satisfaction in eight dimensions: Communication Related to the Task (CRT), Coordination (CO), Definition of Expectations (DE), Knowledge Exchange (KE), Overcoming prejudices (OP), relationship building (RB), trust development (TD), intercultural learning (IL). The questionnaire was first applied in a German online higher education program. After the validation, the questionnaire was translated into Spanish and reviewed by three experts. A final version in Spanish was consolidated after this process. The questionnaire was then applied to the sample of this research and was statistically validated using Confirmatory Factor Analysis (see Appendix C).

Measuring Online CPS Quantity: CPS. The assessment of CPS processes instrument was an own elaboration based on PISA (2017) and evaluated the extent to which specific key group processes were present during the task.⁴ This instrument received the name of "Scale for the assessment of Group Processes in Collaborative Problem Solving" and measures key group processes in four dimensions: Exploring and understanding (EU), Representing and Formulating (RF), Planning and Executing (PE), and Monitoring and Reflecting (MR). The questionnaire was applied to the sample of this research and was statistically validated using Confirmatory Factor Analysis (see Appendix D).

The characteristics of each of the instruments, SGP and CPS, are summarized in Table 2. The psychometric properties of the two scales are detailed in Appendix C and D and show good reliability and appropriate levels for fit indices.

5.4. Analytic strategy

This study uses a variety of methodological approaches to answer the research questions. Methods specific to each research question are described below.

² Including the Netherlands, Spain, and UK in Europe, and Brazil, Uruguay, and Chile in South America (<https://www.lse.ac.uk/media-and-communications/research/research-projects/disto>).

³ The questionnaire underwent a thorough validation process, beginning with expert validation and adaptation through 15 cognitive interviews. A pilot study followed, involving 40 individuals to assess the questionnaire and an additional 200 to pilot the scales. Subsequently, a representative survey was conducted with 1100 individuals in Chile. Additionally, 20 interviews were carried out to assist in interpreting the survey data (<https://www.lse.ac.uk/media-and-communications/research/research-projects/disto/disto-chile>).

⁴ All the fitting indicators are adequate. The article about the construction and validation of this instrument is under review to be published.

Table 1
From digital skills to tangible outcomes questionnaire.

Questionnaire	Number of items	Scale ^a	Factors	Time
DISTO	20	1 = strongly disagree 2 = disagree 3 = neither agree nor disagree 4 = agree 5 = strongly agree	Operational skills Navigation and information skills Social skills Creative skills	10 min
Example items by factor: Operational: “I know how to use shortcut keys (e.g. CTRL-C for copy, CTRL-S for save);” “I know how to bookmark a website”. Navigation and Information: “I find it hard to decide what the best keywords are to use for online searches;” “I find the way in which many websites are designed confusing”. Social: “I know how to change who I share content with (e.g., friends, friends of friends or public);” “I am careful to make my comments and behaviors appropriate to the situation I find myself in online”. Creative: “I know how to make basic changes to the content that others have produced;” “I know how to design a website”.				

^a The original English version uses a self-reported truth scale: responses items ranging from “not at all true of me” to “very true of me.” Here, we present a translation of the Chilean version which uses an agreement scale (Claro, 2017).

Table 2
Satisfaction with group processes and assessment of CPS questionnaires.

Questionnaire	Number of items	Scale	Factors	Time
SGP	24	Regarding the experience with your work group, how satisfied are you with? 1 = very dissatisfied 2 = dissatisfied 3 = indifferent 4 = satisfied 5 = very satisfied	Communication Related to the Task Coordination Definition of Expectations Knowledge Exchange Overcoming Prejudices Building Relationships Trust Development Intercultural Learning	12 min
Example items by factor: Communication Related to the Task: “The way we communicated about the activity”. Coordination: “The way we coordinate for group work”. Definition of Expectations: “The similarity of everyone’s expectations regarding the group work”. Knowledge Exchange: “The extent of knowledge exchange among each individual in the group”. Overcoming Prejudices: “The way we overcame barriers to get to know each other”. Building Relationships: “The way we built a cohesive group”. Trust Development: “The way we overcame difficulties to build mutual trust”. Intercultural Learning: “The way we overcame issues in integrating different viewpoints, perspectives, and ideas”.				
CPS	24	To what extent do the following statements reflect what happened during the task? 1 = not at all 2 = a little 3 = partly 4 = to a great extent 5 = completely	Exploring and understanding Representing and Formulating Planning and Executing Monitoring and Reflecting	12 min
Example items by factor: Exploring and understanding: “We assigned time to get to know each person’s opinions and viewpoints on the matter of the task”. Representing and Formulating: “We made diagrams or drawings that helped us organize the work”. Planning and Executing: “We were able to identify the specific tasks that would be necessary to solve the problem”. Monitoring and Reflecting: “During the task, we reflected on our way of organizing the work, making changes to the strategy when necessary”.				

RQ1. How many distinctive students’ profiles can be identified according to the students’ digital skills? How do these profiles vary across the four digital skills dimensions: Operational, Information Navigation, Social, and Creative?

We conducted a k-means cluster analysis to assess the existence and number of profiles of students in the sample according to their digital skills. K-means clustering is a widely used unsupervised learning algorithm that detects distinctive groups (clusters) within a dataset (Kodinariya & Makwana, 2013; Forgy, 1965).

In this study, each of the four dimensions of the DISTO scale (Operational, Navigation and Information, Social, and Creative) were transformed into z-scores and then used to generate specific digital skills profiles utilizing the k-means. We ran the k-means algorithm over different sets of clusters and then calculated several cluster validity metrics to choose the optimal number of clusters k (see Appendix E).

RQ2. How do students’ socioeconomic backgrounds relate to the previously identified digital skills profiles? Is there evidence of a digital divide?

Using the Chi-Square Test of Independence, we tested the predictive power of socioeconomic characteristics on cluster membership. We conducted the chi-square test of independence to assess the relationship between students' digital skill profiles and a series of socioeconomic characteristics, including sex, age, high school graduation year, school type, parents' level of education, career type and availability of Internet connection and sharing connection device with others.

RQ3. To what extent do digital skills profiles (cluster membership) predict satisfaction with working collaboratively on an online CPS task? Which additional factors influence this satisfaction?

RQ4. To what extent do digital skills profiles (cluster membership) predict the evaluation of the online CPS task? Which additional factors influence this evaluation?

We used multiple linear regression to identify the relationship between digital skills profiles and satisfaction with and evaluating of an online CPS task (Neter et al., 1996). We ran nine regression models to assess the relationship between digital skills profiles and SGP. In each model, the dependent variable is the unstandardized score in one of the eight dimensions of the SGP scale plus an average across dimensions of SGP (CRT=Communication Related to the Task, CO=Coordination, DE=Definition of Expectations, KE=Knowledge Exchange, OP=Overcoming Prejudices, BR=Building Relationships, TD=Trust Development, IL=Intercultural Learning, TOTAL SGP=Average across the eight SGP dimensions).

Similarly, to examine the relationship between digital skills profiles and CPS, we run 5 regression models. In each model, the dependent variable is the unstandardized score in one of the four dimensions of the CPS scale plus an average across dimensions of CPS (EU=Exploring and understanding, RF=Representing and Formulating, PE=Planning and Executing, MR=Monitoring and Reflecting, TOTAL CPS=Average across the four CPS dimensions). The predictor key variable is cluster membership to a digital profile found by applying k-means clustering. In addition, other covariates were added to the models to control for socioeconomic characteristics of the student (e.g., sex, age, high school graduation year).

6. Findings

6.1. Research question 1: identifying students' profiles according to their digital skills

A three-cluster solution was selected based on the suggestion of an optimal number of clusters of different indices (see Appendix E), as well as the interpretability and adequate size of the solution. Each identified cluster was named according to its most salient characteristics regarding the mean scores on the different dimensions of the DISTO scale. Table 3 summarizes each identified cluster's name, description, and size.

The first cluster, high-skilled, is characterized by scores at least 0.5 s.d. above the sample mean in all four dimensions of DISTO. This cluster is the second largest, representing 40.7% of the total sample, with 382 individuals. The second cluster, medium-skilled, is characterized by scores around the sample average (no further than 0.5 s.d. below the mean) in all dimensions of DISTO. This cluster is the largest and comprises 490 individuals, representing 52.2% of the total sample. The third cluster, low-skilled, is the smallest, having 67 students, representing 7.1% of the full sample. It is characterized by scores below the mean in all dimensions of DISTO, except on *Information Navigation*, whose score is very close to the sample average (0.07 s.d. below the mean). The mean z-scores by cluster and DISTO dimension are shown in Table 4 and graphically represented in Fig. 3.

6.2. Research question 2: predictors of cluster/profile membership

Overall, there was a significant relationship between cluster membership and age, father's education, career type, Internet connection availability, and connection device sharing (Table 5). Regarding age ($\chi^2(2) = 10.69$, $p = 0.005$, $V = 0.12$), older individuals (21 years or older) were significantly more likely to belong to the low-skilled cluster. In contrast, younger individuals were significantly more likely to belong to either the high-skilled or the medium-skilled cluster.

A significant relationship was found between the father's education level and cluster membership ($\chi^2(6) = 18.71$, $p = 0.005$, $V = 0.10$). Specifically, students whose fathers' maximum level of education is elementary school were more likely to belong to the low or medium-skilled cluster and less likely to belong to the high-skilled cluster. On the contrary, students whose fathers have at least accessed higher education were more likely to belong to the high-skilled.

Career type was also significantly related to cluster membership ($\chi^2(4) = 9.32$, $p = 0.054$, $V = 0.07$). Students majoring in management, hospitality, and tourism were significantly overrepresented in the high-skilled cluster and underrepresented in the low-skilled cluster. On the other hand, students majoring in health careers were significantly underrepresented in the high-skilled cluster.

Regarding the availability of Internet connection and connection device, there was a significant relationship between having access to Internet at home and cluster membership ($\chi^2(2) = 8.34$, $p = 0.015$, $V = 0.09$). Students with Internet access at home were significantly more likely to belong to the high-skilled cluster and less likely to belong to the medium or low-skilled cluster. In addition, there was a significant relationship between the usual place of Internet connection and cluster membership ($\chi^2(2) = 11.97$, $p = 0.003$, $V = 0.11$). Students frequently connecting from home were more likely to belong to the high-skilled cluster and less likely to belong to the low-skilled one. Lastly, a significant relationship was found between sharing the connection device and cluster membership ($\chi^2(4) = 33.50$, $p = 0.000$, $V = 0.13$). Students whose device is only for personal use were significantly more likely to belong to the high-skilled cluster and less likely to belong to the medium or low-skilled one. Individuals who must share their device with one more person were likelier to belong to the medium-skilled cluster, whereas individuals who share their device with two persons or more

Table 3
Cluster description and size.

Cluster Name	Description	n	% of the sample
High-skilled	Students with skills above average in all dimensions of the DISTO scale	382	40.68
Medium-skilled	Students with skills around average in all dimensions of the DISTO scale	490	52.18
Low-skilled	Students with skills below average in the Operational, Social and Creative dimensions, and around the mean in the Information Navigation dimension	67	7.14

Table 4
Standardized mean of DISTO scale by cluster and dimension.

	High-skilled (n = 382, 40.68%)		Medium-skilled (n = 490, 52.18%)		Low-skilled (n = 67, 7.14%)	
	M	SD	M	SD	M	SD
Digital Skills to Tangible Outcomes (DISTO)						
Operational	0.63	0.47	-0.15	0.64	-2.48	1.06
Information Navigation	0.64	0.91	-0.49	0.77	-0.07	0.94
Social	0.70	0.48	-0.26	0.66	-2.04	1.45
Creative	0.67	0.83	-0.39	0.79	-0.98	0.93

were likelier to belong to the low-skilled cluster.

Cluster membership was not significantly related to sex, graduation year, school type, school schedule, or whether the student works.

6.3. Research question 3: digital skills profiles as predictor of students' satisfaction with group processes

Overall, results suggest that, on average, belonging to the medium or low-skilled cluster predicts a lower satisfaction with the group processes involved in the collaborative online task than belonging to the high-skilled cluster. This result holds, independent of sex, age, and socioeconomic background of the student (see Table 6).

Model 1 indicates that belonging to the low-skilled cluster predicts a significantly lower satisfaction with how communication tasks related to the online task were handled ($B = -0.807$, $t = -1.69$, $p = .091$) compared to the high-skilled cluster. Model 2 suggests that belonging to the medium-skilled cluster predicts a significantly lower satisfaction with coordination tasks while completing the online task than belonging to the high-skilled cluster ($B = -0.421$, $t = -1.77$, $p = .077$). Model 3 finds that membership to either cluster, medium or low-skilled, predicts a significantly lower satisfaction level with the group process of delineating expectations about the online task ($B = -0.591$, $t = -2.46$, $p = .014$; $B = -1.114$, $t = -2.35$, $p = .019$). Model 4 shows that students in the medium-skilled cluster present a significantly lower satisfaction level with the manner and degree of exchanging knowledge within the group while working on the online task, compared to the high-skilled cluster ($B = -0.540$, $t = -2.34$, $p = .019$). Model 5 shows that students from the medium-skilled cluster present a significantly lower level of satisfaction with the way and degree of overcoming prejudices as a group throughout completing the online project concerning the high-skilled cluster ($B = -0.515$, $t = -2.10$, $p = .036$). According to Model 6, belonging to the medium or low-skilled cluster is related to significantly lower satisfaction levels regarding the way and extent of building relationships between group members ($B = -0.602$, $t = -2.38$, $p = 0.018$; $B = -0.884$, $t = -1.77$, $p = 0.077$). Model 7 indicates that students belonging to the medium cluster feel significantly less satisfied with the way of developing and the level of trust within the group ($B = -0.617$, $t = -2.61$, $p = 0.009$). Model 8 suggests that belonging to the medium cluster predicts a significantly lower satisfaction level with the intercultural exchange and learning within the group, compared to the high-skilled cluster ($B = -0.682$, $t = -2.72$, $p = 0.007$).

In addition, models show that sex, year of high school graduation, class schedule, and device sharing are variables significantly related to some of the SGP dimensions. Specifically, being a female predicts significantly higher levels of satisfaction for all factors of SGP scale, while having graduated from high school more recently (in the year 2020) predicts a significantly higher level of satisfaction for the dimensions CO, KE, OP, and RB. On the other hand, having classes in both day and evening schedule is related to a lower level of satisfaction in the dimensions Communication Related to the Task (CRT), Defining Expectations (DE) and Building Relationships (BR). Lastly, students who share their device with two persons or more show significantly higher levels of satisfaction with group processes (in the CO, DE, KE, OP, IL dimensions) compared to students who have personal devices.

6.4. Research question 4: digital skills profiles as predictor of students' assessment of the collaborative problem-solving task

Overall, results suggest that, on average, belonging to the medium or low-skilled cluster predicts a poorer evaluation of the group processes involved in the collaborative online task than belonging to the high-skilled cluster. This result holds, independent of sex, age, and socioeconomic background of the student (see Table 7).

Table 5
Socioeconomic characteristics of the clusters.

	High- skilled (n = 382, 40.68%) %	Medium- skilled (n = 490, 52.18%) %	Low- skilled (n = 67, 7.14%) %	Total (n = 939) %	χ^2 (df)	p
Sex						
Female	54.7	58.0	49.3	56.0	4.47 (4)	0.347
Male	44.5	40.8	47.8	42.8		
Other	0.8	1.2	3.0	1.2		
Age (years)						
Between 18 and 20	47.6	49.6	28.4	47.3	10.69 (2)	0.005
21 or older	52.4	50.4	71.6	52.7		
High school graduation year						
2018 or before	48.7	49.6	64.2	50.3	6.06 (4)	0.195
2019	23.3	22.7	13.4	22.3		
2020	28.0	27.8	22.4	27.5		
School type						
General track	50.8	46.5	38.8	47.7	5.53 (4)	0.237
Vocational-Technical track	42.2	46.7	49.3	45.1		
Other	7.1	6.7	11.9	7.2		
Father's education level						
Primary	21.5	28.6	26.9	25.6	18.71 (6)	0.005
Secondary	47.6	47.8	38.8	47.1		
Higher education	17.0	12.0	9.0	13.8		
Does not know	13.9	11.6	25.4	13.5		
Mother's education level						
Primary	22.0	26.3	35.8	25.2	9.47 (6)	0.149
Secondary	52.1	50.8	41.8	50.7		
Higher education	21.5	19.2	14.9	19.8		
Does not know	4.5	3.7	7.5	4.3		
Career type						
Engineering	40.3	38.8	43.3	39.7	9.32 (4)	0.054
Management, hospitality and tourism	22.5	18.0	9.0	19.2		
Health	37.2	43.3	47.8	41.1		
School schedule						
Day	74.9	75.1	70.2	74.7	4.51 (4)	0.341
Evening	19.6	21.6	26.9	21.2		
Both	5.5	3.3	3.0	4.2		
Do you work?						
Yes	30.6	29.6	34.3	30.4	0.65 (2)	0.723
Do you have internet connection at home?						
Yes	94.5	89.8	86.6	91.5	8.34 (2)	0.015
Usual place of connection						
House	97.4	94.7	88.1	95.3	11.97 (2)	0.003
Other	2.6	5.3	11.9	4.7		
Do you share your usual connection device?						
No	73.6	59.6	46.3	64.3	33.50 (4)	0.000
Yes, with one more person	18.1	27.6	28.4	23.8		
Yes, with two or more persons	8.4	12.9	25.4	11.9		

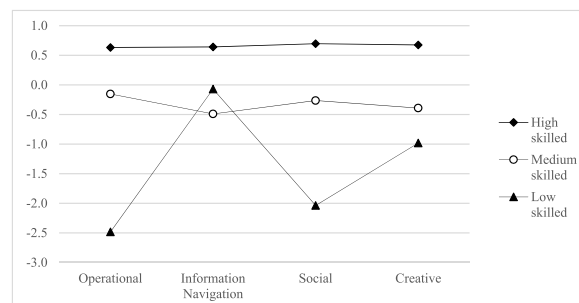


Fig. 3. Digital skills profiles.

Model 1 indicates that belonging to the medium-skilled cluster predicts a significantly lower evaluation of the processes of Exploring and Understanding the online task compared to the high-skilled cluster ($B = -0.932$, $t = -1.72$, $p = 0.086$). Model 2 suggests that belonging to the medium or the low-skilled cluster predicts a significantly lower evaluation of the processes of Representing and Formulating while completing the online task, compared to belonging to the high-skilled cluster ($B = -1.487$, $t = -2.45$, p

Table 6
Digital skills profiles as predictor of students' satisfaction with group processes.

	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6	Model 7	Model 8	Model 9
	CRT	CO	DE	KE	OP	BR	TD	IL	Total SGP
Cluster (base = high- skilled)									
medium-skilled	−0.275 (0.242)	−0.421* (0.238)	−0.591** (0.240)	−0.540** (0.231)	−0.515** (0.245)	−0.602** (0.253)	−0.617*** (0.236)	−0.682*** (0.251)	−0.530** (0.225)
low-skilled	−0.807* (0.477)	−0.541 (0.469)	−1.114** (0.473)	−0.568 (0.454)	−0.732 (0.482)	−0.884* (0.499)	−0.239 (0.465)	−0.747 (0.495)	−0.704 (0.443)
Sex (base = male)									
female	0.818*** (0.300)	0.853*** (0.295)	0.776*** (0.298)	0.814*** (0.286)	0.805*** (0.303)	0.640** (0.314)	0.774*** (0.293)	0.652** (0.311)	0.766*** (0.279)
Age	0.081 (0.125)	0.066 (0.123)	0.009 (0.125)	0.028 (0.120)	0.013 (0.127)	0.066 (0.131)	0.035 (0.122)	0.006 (0.130)	0.038 (0.116)
High school graduation year (base = 2018 or before)									
2019	0.589* (0.352)	0.280 (0.346)	0.364 (0.350)	0.455 (0.336)	0.553 (0.356)	0.422 (0.368)	0.200 (0.344)	0.113 (0.366)	0.372 (0.327)
2020	0.587 (0.360)	0.754** (0.354)	0.507 (0.358)	0.588* (0.343)	0.808** (0.364)	0.710* (0.377)	0.471 (0.351)	0.410 (0.374)	0.604* (0.334)
Career type (base = engineering)									
Administration	−0.294 (0.365)	−0.114 (0.359)	−0.179 (0.363)	−0.457 (0.348)	0.025 (0.369)	0.240 (0.382)	−0.227 (0.356)	−0.121 (0.379)	−0.141 (0.339)
Health	−0.038 (0.342)	0.110 (0.336)	0.143 (0.340)	−0.047 (0.326)	0.331 (0.346)	0.516 (0.358)	0.291 (0.334)	0.303 (0.355)	0.201 (0.317)
Class Schedule (base = day)									
evening	−0.035 (0.337)	−0.199 (0.331)	−0.102 (0.335)	−0.053 (0.321)	0.112 (0.341)	0.232 (0.353)	−0.334 (0.329)	−0.171 (0.350)	−0.069 (0.313)
both	−1.106* (0.582)	−0.890 (0.572)	−1.040* (0.577)	−0.683 (0.554)	−0.928 (0.588)	−1.080* (0.608)	−0.562 (0.567)	−0.854 (0.604)	−0.893* (0.540)
Work (base = do not work)									
Yes	−0.013 (0.291)	−0.044 (0.286)	0.039 (0.289)	−0.127 (0.277)	−0.072 (0.294)	−0.153 (0.305)	−0.140 (0.284)	−0.223 (0.302)	−0.092 (0.270)
Internet connection at home (base = no)									
Yes	−0.067 (0.446)	0.230 (0.439)	0.281 (0.443)	−0.361 (0.425)	0.026 (0.451)	0.050 (0.467)	0.339 (0.435)	0.212 (0.463)	0.089 (0.414)
Place of usual connection (base = other)									
Home	0.454 (0.590)	−0.283 (0.580)	−0.401 (0.586)	0.104 (0.563)	0.337 (0.597)	0.305 (0.618)	0.297 (0.576)	0.373 (0.613)	0.148 (0.548)
Device sharing (base = no)									
Yes, with one other person	0.195 (0.278)	0.165 (0.273)	0.100 (0.276)	−0.047 (0.265)	0.309 (0.281)	−0.011 (0.291)	0.167 (0.271)	0.187 (0.288)	0.133 (0.258)
Yes, with two or more persons	0.540 (0.369)	0.646* (0.363)	0.650* (0.367)	0.783** (0.352)	0.670* (0.373)	0.566 (0.386)	0.554 (0.360)	0.741* (0.383)	0.644* (0.343)
Observations	939	939	939	939	939	939	939	939	939
R-squared	0.032	0.038	0.038	0.038	0.045	0.041	0.046	0.038	0.042

Note. Each column represents the results of one multiple linear regression model, in which the dependent variable is the unstandardized score in one dimension of the SGP scale plus an average across all dimensions of SGP. CRT=Communication related to the Task, CO=Coordination, DE=Definition of Expectations, KE=Knowledge Exchange, OP=Overcoming prejudices, BR=Building Relationships, TD=Trust Development, IL= Intercultural Learning, TOTAL SGP=Average of CRT, CO, DE, KE, OP, RB, TD y IL. Standard error in parentheses; * $p < .1$, ** $p < .05$, *** $p < .01$.

= 0.015; $B = -2.038$, $t = -1.70$, $p = 0.089$). Model 3 finds that membership to the medium-skilled cluster predicts a significantly lower evaluation of the process of Planning and Executing the online task ($B = -1.337$, $t = -2.37$, $p = 0.018$). Lastly, Model 4 shows that students in the medium-skilled cluster show a significantly lower evaluation of the process of Monitoring and Reflecting on the online task compared to the high-skilled cluster ($B = -1.3$, $t = -2.79$, $p = 0.005$).

In addition, models show that sex, year of high school graduation, class schedule, and sharing the device are variables significantly related to some of the CPS scale dimensions. Specifically, being a female predicts significantly higher evaluation scores for all dimensions of the CPS scale, while having graduated from high school more recently (in the year 2020) predicts significantly higher evaluation scores in the factors EU y PE. On the other hand, having classes in both day and evening schedule is related to significantly lower evaluation scores in the EU factor. Lastly, students who share their device with two persons or more, show significantly higher evaluation scores in all the dimensions of the CPS scale compared to students who have a device for personal use.

7. Discussion

Our cluster analysis revealed that there is not a unique student profile among TVHE students in Chile; conversely, we found three substantially distinctive subgroups: the high, medium, and low-skilled cluster (H1). This result exposed considerable heterogeneity among TVHE students' digital skills and socioeconomic backgrounds, factors that proved to be related to their evaluation of and satisfaction with an online CPS task.

Table 7

Digital skills profiles as predictor of students' assessment of the collaborative problem-solving task.

	Model 1	Model 2	Model 3	Model 4	Model 5
	EU	RF	PE	MR	Total CPS
Cluster (base = high-skilled)					
medium-skilled	−0.932* (0.542)	−1.487** (0.608)	−1.337** (0.563)	−1.300*** (0.466)	−1.264** (0.527)
low-skilled	−1.518 (1.068)	−2.038* (1.197)	−1.807 (1.108)	−0.986 (0.917)	−1.587 (1.038)
Sex (base = male)					
female	1.913*** (0.672)	2.284*** (0.754)	1.803*** (0.698)	1.426** (0.577)	1.856*** (0.654)
Age	0.164 (0.281)	0.069 (0.315)	0.138 (0.292)	0.041 (0.241)	0.103 (0.273)
High school graduation year (base = 2018 or before)					
2019	0.944 (0.789)	0.968 (0.884)	1.044 (0.819)	0.314 (0.678)	0.817 (0.767)
2020	1.480* (0.807)	1.452 (0.905)	1.794** (0.838)	0.881 (0.693)	1.402* (0.785)
High school type					
General track	−0.305 (0.545)	−0.206 (0.611)	−0.180 (0.566)	−0.092 (0.468)	−0.196 (0.530)
Vocational/Technical	−0.230 (1.028)	−0.249 (1.153)	−0.293 (1.067)	−0.187 (0.883)	−0.240 (1.000)
Father's education (base = elementary)					
High school	−0.070 (0.636)	0.193 (0.713)	−0.130 (0.660)	−0.326 (0.546)	−0.083 (0.619)
Higher education	−0.491 (0.876)	−0.577 (0.982)	−1.062 (0.909)	−0.906 (0.752)	−0.759 (0.852)
Career type (base = engineering)					
Administration	−0.547 (0.818)	−0.862 (0.917)	0.018 (0.849)	−0.347 (0.703)	−0.435 (0.796)
Health	0.163 (0.766)	0.136 (0.859)	0.790 (0.795)	0.594 (0.658)	0.421 (0.745)
Work schedule (base = day)					
evening	−0.192 (0.755)	−0.121 (0.846)	0.273 (0.784)	−0.505 (0.648)	−0.136 (0.734)
both	−2.292* (1.302)	−2.009 (1.460)	−2.203 (1.352)	−1.415 (1.119)	−1.980 (1.267)
Work (base = do not work)					
Yes	−0.136 (0.652)	−0.117 (0.731)	−0.260 (0.677)	−0.363 (0.560)	−0.219 (0.634)
Internet connection at home (base = no)					
Yes	0.257 (0.999)	0.040 (1.120)	0.174 (1.037)	0.551 (0.858)	0.255 (0.972)
Place of usual connection (base = other)					
Home	0.338 (1.322)	−0.357 (1.482)	0.455 (1.372)	0.669 (1.136)	0.276 (1.286)
Device sharing (base = no)					
Yes, with one other person	0.420 (0.622)	0.135 (0.697)	0.350 (0.646)	0.354 (0.534)	0.315 (0.605)
Yes, with two or more persons	1.423* (0.827)	1.931** (0.927)	1.433* (0.858)	1.295* (0.710)	1.520* (0.804)
Observations	939	939	939	939	939
R-squared	0.036	0.039	0.042	0.045	0.041

Note. Each column represents the results of one multiple linear regression model, in which the dependent variable is the unstandardized score in one dimension of the CPS scale plus an average across all dimensions of CPS. EU=Exploring and Understanding, RF=Representing and Formulating, PE=Planning and Executing, MR=Monitoring and Reflecting, TOTAL CPS=Average of EU, RF, PE, and MR. Standard error in parentheses; *p < .1, **p < .05, ***p < .01.

Our study identified a clear relationship between technology access, digital skill levels, and students' socioeconomic backgrounds (H2). This finding is in line with prior research conducted in different contexts, further confirming the strong link between digital skills and socioeconomic background (e.g., Elena-Bucea et al., 2021; Liu, 2021; Van De Werfhorst et al., 2022). In our study, students belonging to the low-skilled profile tended to be older and have less educated parents. At the same time, a higher proportion of students in this cluster reported having suboptimal conditions for online work (for example, 25% must share their primary connection device with at least two more people versus 8% in the high-skilled cluster).

Among the students, those falling into the lower-skilled profile presented the lowest scores across most DISTO dimensions; namely, Operational, Social, and Creative. This observation resonates with prior research, underscoring the persistence and potential amplification of structural inequalities in the digital world (Cabello & Claro, 2017; Claro et al., 2015; Van Deursen & Van Dijk, 2014, 2019;

Van Dijk, 2013, 2017). Our study revealed that these disparities persist even within the specific context of TVHE in a developing country.

It is worth noting the distribution of digital skills across the four analyzed DISTO dimensions in light of the three clusters profiles: The higher and medium-skilled clusters exhibited a consistent distribution of skills across these dimensions, with the higher-skilled cluster presenting above-average scores in each DISTO dimension, and the medium-skilled cluster showing scores close to the average across all dimensions. In contrast, the low-skilled cluster presented a particular distribution across dimensions: scores similar to the medium cluster in the Information Navigation and Creative dimensions but considerable lower scores in the Operational and Social dimensions. Hence, an alternative perspective could be to consider the lower-skilled cluster as a subgroup of the medium-skilled cluster, facing additional difficulties in particular areas of digital skills related to the operational and social dimensions.⁵ Although, a more in-depth analysis is required, existing literature can offer some insights into the distribution of digital skills across dimensions. According to this, operational skills are more prone to quick changes than the rest of the dimensions because of the rapid and constant development of operational tools (Ala-Mutka, 2011; Iordache et al., 2017). This makes operational skills particularly challenging for the lower-skill group, which is composed of older individuals who may struggle to keep pace with the operational advances (only 28% of students in the low-skilled cluster is 20 or younger, while half of students in the medium cluster are 20 or younger). Social skills, on the other hand, are likely influenced by individual interests, experiences, and opportunities for social interaction (Ferrari, 2013; Hargittai, 2007; Iordache et al., 2017). This skills can be particularly difficult to acquire by the lower-skilled cluster because of their more limited availability of time (e.g., 34% reports to work, compared to 30% in the medium-skilled cluster) and digital resources (e.g., 25% shares the usual connection device with two or more persons, compared to 13% of the medium-skilled cluster); thus, affecting their possibility of creating an online identity and engage with others online.

Our results also shed light on a relationship that has been scarcely examined -and even less studied in the context of TVHE: the link between digital skills and CPS. With help of the DISTO framework (Helsper et al., 2015), we conceptualized satisfaction with and evaluation of the CPS task as a tangible outcome from the economic-education subfield that, in this case, is beneficial for developing students' educational path in TVHE. This is because completing this online CPS task is an important milestone for TVHE students to accomplish a first-year class successfully. According to this conceptual model, lower levels of digital skills lead to lower levels of engagement with ICTs, which in turn results in a lower likelihood for an individual to achieve tangible outcomes (Van Deursen et al., 2017). Our results empirically confirmed this relationship in the TVHE context revealing that on average, belonging to the medium or low-skilled cluster predicted a lower satisfaction with and a poorer evaluation of the group processes involved in the collaborative online task (H3.1 and H4.1). Additionally, some specific socioeconomic background factors, such as gender and having to share a device influence satisfaction and evaluations (H3.2 and H4.2).

Consequently, the empirical evidence underscores how deficiencies in digital competencies and offline available resources, as highlighted by the model, not only impact online engagements but also reverberate into the realm of online CPS, influencing students' perception and assessment of their educational experiences within the TVHE landscape. The alignment between the conceptual framework and our empirical findings reinforces the intricate connection between digital skills and tangible educational outcomes. This substantiates the conceptual model's relevance for the realms of TVHE and CPS, an intersection that has not been previously examined using this lens. Our study also highlights the direct and strong connection between the online and offline spheres in the corresponding fields model within the economic-educational dimension: online educational work on the CPS task (online world) translates into a satisfactory experience and a better evaluation of a concrete educational task which is required to further develop the student educational path in TVHE (offline world).

All in all, we found that within the group of TVHE students' those who suffer a combination of social disadvantages such as lower socioeconomic background, less than optimal access and conditions for online work, and inferior digital skills, are the most challenged to be able to obtain an actual benefit from the use of ICTs. This result resonates with previous findings regarding the digital divide, which state that technology can amplify the effect of other drivers of inequality because what individuals can get out of technology depends on their initial circumstances and context (e.g., Elena-Bucea et al., 2021; Robinson, 2009; Robinson et al., 2015).

7.1. Policy implications for TVHE

Social policy has a crucial role in addressing digital inequalities and their effects. Providing quality access and promoting the use and engagement with technology is a start; nevertheless, as our results suggest, this action on its own will certainly not be enough to close the digital gap due to the multidimensional nature of the problem. Next, we outline some strategies for consideration by education and training institutions related with a targeted and tailored support for the most disadvantaged profile.

The Chilean school system is characterized as highly segmented by socioeconomic class at all levels (primary, secondary, and higher education) (Dupriez et al., 2023) and TVHE is no exception. However, we could identify significant differences even among students from the same institution and cohort. We found that 7% of students endured a combination of social and digital disadvantages, translating into a lower evaluation and satisfaction with an online CPS task. This vulnerable group of students needs to be focalized, tailored, and receive multidimensional support.

To start, quality access to ICTs and improvement of the current conditions for working online show to be essential for the most disadvantaged profile. Specifically, ensuring quality Internet access at home and a device for personal use could even up the field at a

⁵ Note that a two-cluster solution for the optimal number of clusters was suggested by some of the fit indices calculated in the cluster analysis (See Appendix E).

foundational level. Equalizing conditions for optimal online work is the first step to addressing the digital divide.

Within the digital arena, it is crucial to compensate and balance up the digital skills of the low and medium-skill clusters, specially, in the Operational dimension for the low-skill cluster. This is a foundational ability that may be hindering the development of more sophisticated ones, such as the Social and Creative. For this, TVHE education institutions are encouraged to: i) develop a systematic strategy to address the inequalities in digital skills among their students, starting by identifying specific digital skill fields that need to be improved; and ii) provide short courses or modules targeted to low and medium-skilled individuals, to level up specific digital skills. Other measures that educational institutions can take are to invest in platforms and digital learning environments that are intuitive and easy to handle for all students to make the user self-sufficient and, at the same time, provide students with quality and timely technical support.

The analysis empirically showed the links between social and digital disadvantage and the connection between digital disadvantage and challenges to translate technology engagement into a tangible educational outcome, namely, satisfaction with and self-evaluation of an online CPS task. Both CPS and digital skills are central to the 21st-century skill set, and TVHE institutions have a crucial role in developing these skills for all students, carefully ensuring to not leave behind those on the margins: the socially, economically, and digitally disadvantaged (Helsper, 2008, 2015; Van Deursen et al., 2017).

7.2. Limitations

One of the primary limitations of our study lies in the use of a self-reported instrument to measure digital skills. Self-reported measures can be subject to biases such as social desirability and individual self-beliefs (Rosman et al., 2015). As a result, these measures may not fully capture the comprehensive spectrum of participants' digital skills. Still, we used DISTO, a self-reported questionnaire that is aligned with our conceptual framework, has been extensively and rigorously validated in several contexts, including Chile, and that reported good internal consistency and fit within our study sample. Future studies could enhance methodological robustness by incorporating performance rather than self-report assessments of digital skills to ensure a more accurate representation of participants' digital proficiency.

Second, although the teacher leading the classroom explained and highlighted the collaborative nature of the task, the design of the CPS task itself did not have an explicit mechanism for ensuring interdependence among participants and their roles. The absence of interdependence might have influenced students' engagement levels and their perceived satisfaction and evaluation of the task. Literature underscores the significance of interdependent tasks in eliciting deeper collaboration among participants (Scoular et al., 2020). Introducing CPS tasks that emphasize interdependence could potentially yield different insights into how digital skills and socioeconomic factors intersect with CPS related outcomes. Future research should explore and incorporate interdependence into the design of the CPS tasks explicitly in order to explore its effects on students' engagement and satisfaction.

Despite these limitations our study offers valuable insights into the interconnections between digital skills, socioeconomic background, and online CPS tasks within an understudied context: technical and vocational higher education.

7.3. Future directions

The landscape of digital skills and online CPS is continually evolving, influenced by emerging ICTs such as Artificial Intelligence (AI) and its dynamic capabilities. Understanding the implications of these novel technologies presents a promising avenue for future research.

AI's versatile nature and its accelerated use in educational settings raises intriguing questions about its role in shaping the trend of the digital divide as well as the designs, dynamics, and monitoring processes of CPS. Research could focus, for instance, on the integration of AI algorithms within CPS frameworks and examine how AI influences task complexity, decision-making processes, and group dynamics. Understanding the interplay between human cognition, digital skills, and AI could illuminate new avenues for optimizing CPS tasks and methodologies.

Declaration of competing interests

The authors declare that they have no competing interests.

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CRedit authorship contribution statement

Daniela Luengo-Aravena: Writing – review & editing, Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Patricio Cabello:** Writing – review & editing, Supervision, Investigation, Conceptualization. **Beatriz Rodriguez-Milhomens Bachino:** Writing – review & editing, Investigation.

Data availability

The authors do not have permission to share data.

Appendix

Appendix A. Task details and material

The Task: Global Warming and Consequences for Life

For several years, experts have warned about global warming and the consequences it is having and will have on nature and human life. This has given rise to a large amount of information circulating in the media such as television, radio, the press and social networks.

One day, going through Instagram, you find a post where a highly respected French economist says: “human life is destined for extinction, since it is not possible to generate economic development without destroying the planet; therefore, extinction is a matter of time. There is nothing we can do”. Once you read it, you feel like you would like to say something about it.

That afternoon you meet your friends and since you kept thinking about it, you show them the economist’s publication. Suddenly, a colleague discovers that the same publication by the economist has been commented on in an electronic newspaper with a message that says: “send us your comment on this article and we will publish it”. The group then decides that they will write a note together to send to the journal.

Note: The newspaper accepts texts from readers as long as they are between 500 and 800 words.

Now comment on the news! Before answering, reflect as a group to send a text that represents all of you.

General guidelines for introducing the CPS task to the students

This CPS task is a training task whose objective is the elaboration of an opinion text, elaborated collectively, in relation to a contentious subject. For this, students are assigned to random groups and are asked to work, over a two weeks period, on a relevant and current issue.

Students are expected to prepare the text in response to a press release considering:

1. An autonomous work, with little teacher mediation
2. A collaborative work, where they can resolve eventual conflicts and reflect points of view
3. A synthesis work, using a written text and an oral synthesis, that reflects the central argument and the way in which they arrived at it.

Specific guidelines for the development of the task

Week	Instructions
Week 10	» The teacher forms the work groups RANDOMLY in the virtual classroom. If students object, they should be told that working with people they may not know as well is part of the job. » The groups must be made up of 4 or 5 people. » The general guidelines and then the actual problem is delivered to the students. They meet with their groups in the virtual spaces created, exchange contacts and plan the start of collaborative work (20 min). » Then, students, individually, answer a general characterization survey (25 min) that will be housed in her learning environment. » The teacher closes the session by answering the questions of the students (5 min).
Week 11	» The teacher provides usual instruction -not related to the CPS task that is being implemented (30 min) » At the end of the class (last 20 min), the teacher asks: a) have you progressed with the group task? b) have you understood all the instructions? c) Have you had any difficulties? » The teacher must avoid influencing the development of the task itself, indicating that they must resolve all situations of both the task and the functioning of the group as a group. It should be emphasized that group work difficulties are situations that frequently occur in working life and that it is important to train to resolve them. » Students are reminded that the deadline of the task is next session, and that the group work must be uploaded by then.
Week 12	» Before the start of the class, the students upload the solution to the problem to the platform. The teacher must remind them at the beginning of the class, indicating that, if they have not done that, they should do it at that moment. » The teacher asks the students of each group to briefly comment on how they developed the group work, what was the central argument of the discussion and what were the steps to arrive at the elaborated text. (20 min). » Students must be asked to answer a final questionnaire, which will be answered in the learning environment (20 min). The students are told that once the questionnaire is finished, the task will be closed. » Final conclusions of the task (10 min). The teacher emphasizes the importance of the task, highlighting the interesting things that are learned by working in a group with people with whom one has not necessarily worked before.

Appendix B. Fit indices and descriptive statistics, DISTO

Table B.1

CFA fit indices DISTO (n = 939).

	$\chi^2/g.l$	P	RMSEA (IC 95%)	CFI	TLI
4 Factors, 20 ítems	6.70	<0.001	0.078 (0.074–0.082)	0.914	0.901
Reference values	<2	>0.05	<0.08 adequate <0.05 optimum	>0.90 adequate >0.95 optimum	>0.90 adequate >0.95 optimum

Table B.2

Internal consistency, DISTO (n = 939).

Dimension	Number of items	Cronbach's Alpha	Ordinal Alpha
Operational	5	0.868	0.925
Navegation and Information	5	0.773	0.765
Social	5	0.856	0.901
Creative	5	0.730	0.718

Table B.3

Descriptive statistic, DISTO (n = 939).

	Operational	Navegation Information	Social	Creative	Total
Mean	21.1	13.7	21.3	15.2	17.8
Standard Dev.	4,1	3.9	3.5	4.0	2.3
Min	5	5	5	5	5
Max	25	25	25	25	25
Standardized asymmetry coef.	−21.0	−0.1	−18.5	−0.5	−13.5

Appendix C. Fit indices and descriptive statistics, SGP

Table C.1

CFA fit indices, SGP (n = 939).

	$\chi^2/g.l$	P	RMSEA (IC 95%)	CFI	TLI
8 factors, 24 ítems	5.23	<0.001	0.067 (0.063–0.071)	0.996	0.996
Reference values	<2	>0.05	<0.08 adequate <0.05 optimum	>0.90 adequate >0.95 optimum	>0.90 adequate >0.95 optimum

Table C.2

Internal consistency, SGP (n = 939).

Factor	Number of items	Cronbach's Alpha	Ordinal Alpha
Communication Related to the Task	3	0.944	0.963
Coordination	3	0.949	0.969
Definition of Expectations	3	0.950	0.968
Knowledge Exchange	3	0.962	0.974
Overcoming Prejudices	3	0.956	0.970
Building Relationships	3	0.962	0.979
Trust Development	3	0.968	0.978
Intercultural Learning	3	0.965	0.977

Table C.3

Descriptive statistics, SGP (n = 939).

	CRT	CO	DE	KE	OP	BR	TD	IL	Total
Mean	16.8	17.3	17.0	17.0	17.8	17.5	16.3	16.8	17.1
Standard Dev.	3.3	3.7	3.5	3.7	3.1	3.3	4.0	3.9	3.0
Min	3	3	3	3	3	3	3	3	3
Max	15	15	15	15	15	15	15	15	15
Standardized asymmetry coef.	−11.0	−11.5	−11.7	−9.9	−13.0	−11.1	−13.7	−10.0	−13.0

CRT=Communication Related to the Task, CO=Coordination, DE = Definition of Expectations, KE=Knowledge Exchange, OP= Overcoming Prejudices, BR= Building Relationships, TD = Trust Development, IL= Intercultural Learning, TOTAL = Average of the eight SGP dimensions.

Appendix D. Fit indices and descriptive statistics, CPS

Table D.1

CFA fit indices, CPS (n = 939).

	$\chi^2/\text{g.l}$	P	RMSEA (IC 95%)	CFI	TLI
4 factors, 24 items	5.60	<0.001	0.070 (0.066–0.074)	0.988	0.986
Reference values	<2	>0.05	<0.08 adequate <0.05 optimum	>0.90 adequate >0.95 optimum	>0.90 adequate >0.95 optimum

Table D.2

Internal consistency, CPS (n = 939).

Factor	Number of items	Cronbach's Alpha	Ordinal Alpha
Exploring and Understanding	6	0.943	0.955
Representing and Formulating	6	0.948	0.961
Planning and Executing	6	0.958	0.969
Monitoring and Reflecting	6	0.926	0.946

Table D.3

Descriptive statistics, CPS (n = 939).

	EU	RF	PE	MR	Total CPS
Mean	22.23	22.82	22.60	22.34	22.50
Standard Dev.	6.69	6.71	7.03	6.75	6.55
Min	6	6	6	6	6
Max	30	30	30	30	30
Standardized asymmetry coef.	−11.91	−13.58	−12.48	−12.33	−13.44

EU = Exploring and Understanding, RF=Representing and Formulating, PE=Planning and Executing, MR=Monitoring and Reflecting.

Appendix E. Optimal number of clusters

Table E.1

Results according to different clustering methods.

Clustering Method	(1) Optimal number of cluster	(2) Indices	(3) Within_SS/Total_SS
k-means	3	8	0.730
Ward	3	7	0.741
Ward 2	3	10	0.759
Centroid	3	9	0.775
Single	3	7	0.775
Complete	2	11	0.774
Median	2	10	0.775
McQuitty	2	13	0.774
Average	4	7	0.774

Notes. Columns show: (1) the optimal number of clusters as suggested by the majority of indices; (2) number of indices proposing the optimal number of clusters. For each clustering method, 24 indices were calculated to evaluate the optimal number of clusters⁶; (3) proportion of the within-cluster sum of squares out of the total sum of squares.

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⁶ Using the NbClust procedure in R, 24 indices were calculated to evaluate the optimal number of clusters (see Charrad et al., 2014; for further detail).

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